

Aletheia

AI Business Knowledge Assistant

Ask a business question in plain language. It routes to the right source, answers with hybrid SQL + document retrieval, and attaches a citation to every fact — one you can trace to the exact database row or document page.

► [Try the live demo](#) · [View the source](#)

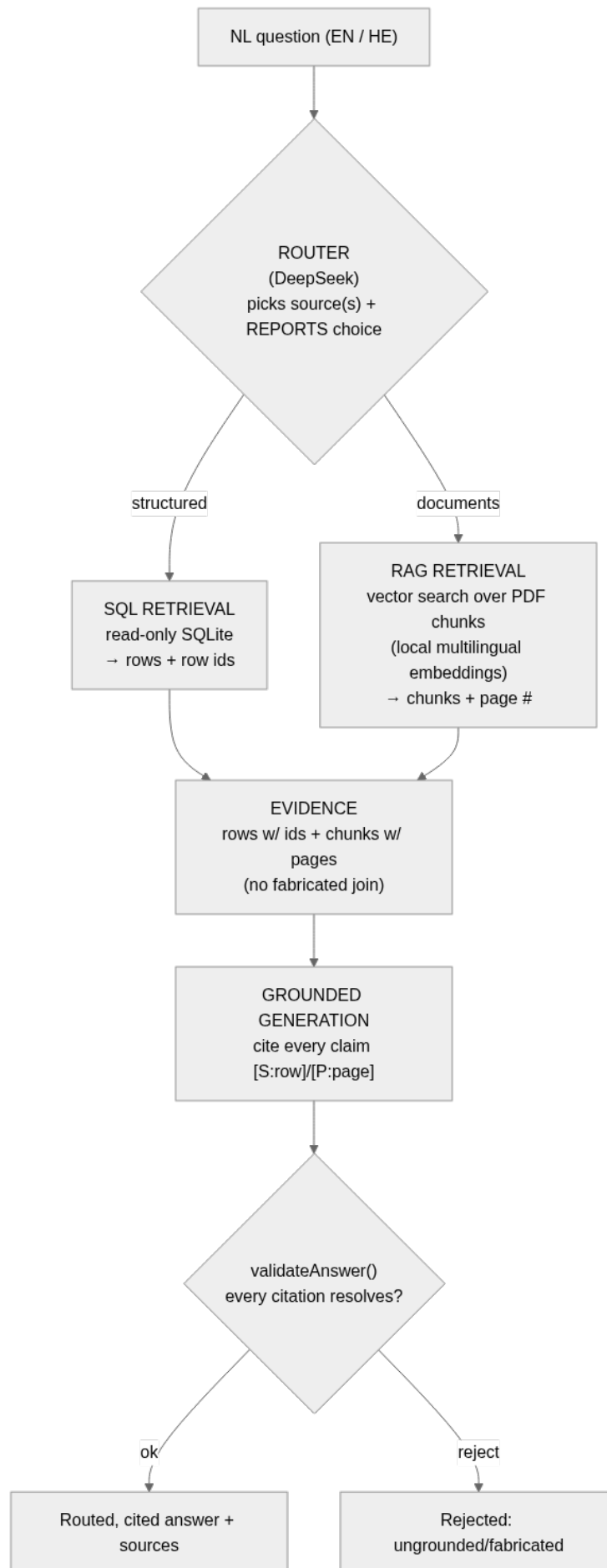
query routing · hybrid SQL + RAG · grounded generation · source attribution · English + Hebrew

This is not a PDF chatbot, and not "upload PDFs into a vector database and do semantic search" — the approach you explicitly ruled out. It is a real **query-routing → hybrid-retrieval → grounded-generation → source-attribution** pipeline, built to extend to your future CRM / email / cloud-storage / case-management sources.

At a glance

What it does	Routes each question to the relevant source(s), retrieves with SQL + RAG, returns a cited answer
The differentiator	Honest grounding — every claim cites a resolvable source, or it says the data doesn't have it
Capabilities	Contract Intelligence · Case File Q&A · Maintenance Spend Intelligence
Languages	English + Hebrew (live today)
Stack	Next.js · DeepSeek · local multilingual embeddings · bundled SQLite + vector index · Vercel
Quality floor	40 unit + 13 journey tests (run against the live deploy) · CI-enforced

How it works



Architecture: routing → hybrid retrieval → grounded generation → validation

- 1. Query routing** — a real LLM decision picks the relevant source(s) and **reports it** in the UI (it's not a hidden heuristic).
- 2. Hybrid retrieval** — SQL over your structured data (every row carries a stable id) + RAG over your documents (every chunk carries its page).
- 3. Grounded generation** — the answer is composed **only** from retrieved evidence, with an inline citation on every fact.
- 4. Source attribution + a content-fidelity gate** — `validateAnswer()` rejects any answer whose citation doesn't resolve to real evidence, so "fluent but ungrounded" can't ship. The two source domains (structured vs. documents) are **never merged on an invented join**.

Full architecture + the data-flow diagram: [architecture.md](#). Engine internals: [features/shared-engine/reference.md](#).

What it can do (three worked capabilities)

Each capability is a real feature with its own golden examples, regression tests, and the screenshots below — all captured from the live demo.

1. Contract Intelligence — expiry, value, and honest about penalties

Ask "What contracts expire in the next 90 days and what penalties are defined in those contracts?" → **38 contracts** expiring, combined annual value **\$18,924,883.79**, each row cited — and an **honest statement that penalty terms are not available** (your contract data has no penalty field and no contract documents). It never fabricates a penalty and never pulls from the unrelated case file.

Ask a business question in plain language. Aletheia routes it to the right source, answers with hybrid SQL + document retrieval, and attaches a citation to every fact you can trace to the exact row or page. Not a PDF chatbot — grounded, or it says so.

What contracts expire in the next 90 days and what penalties are defined in those contracts?

ASK

ROUTING DECISION

question →

StructuredSQL

Contracts expiring is a structured query; penalties would be in contract documents, but no such documents exist in this system. (contract questions are answered from structured data only; no contract documents exist in this corpus, so penalty terms are reported as unavailable rather than sourced from an unrelated document)

Intents: contracts_expiring

ANSWER

There are **38 contracts** expiring in the next 90 days (between 2026-06-09 and 2026-09-07), with a combined annual value of **\$18,924,883.79** (verified SQL aggregate over the cited rows). Representative examples include:

- Dental Hygienist (Edgepulse), ends 2026-06-11, annual cost \$779,823.65 [S:contracts#760]

- Paralegal (Voomm), ends 2026-06-11, annual cost \$95,103.45 [S:contracts#737]

- Geological Engineer (Realbuzz), ends 2026-06-12, annual cost \$133,353.76 [S:contracts#523]

- Quality Engineer (Brainsphere), ends 2026-06-17, annual cost \$844,932.35 [S:contracts#999]

- Occupational Therapist (Fanoodle), ends 2026-06-17, annual cost \$132,126.71 [S:contracts#673]

- Staff Scientist (Feedfish), ends 2026-06-17, annual cost \$890,633.77 [S:contracts#393]

- VP Quality Control (Topicware), ends 2026-06-17, annual cost \$673,961.67 [S:contracts#713]

- Office Assistant IV (Dazzlesphere), ends 2026-06-18, annual cost \$769,569.40 [S:contracts#544]

- Automation Specialist IV (Zoombeat), ends 2026-06-21, annual cost \$428,720.60 [S:contracts#29]

- Sales Representative (Chatterbridge), ends 2026-06-27, annual cost \$672,942.87 [S:contracts#476]

- Teacher (Demimbu), ends 2026-06-30, annual cost \$170,152.10 [S:contracts#43]

- Research Associate (Zoomcast), ends 2026-07-02, annual cost \$276,861.23 [S:contracts#167]

- Graphic Designer (Skivee), ends 2026-07-06, annual cost \$333,924.27 [S:contracts#563]

- Project Manager (Skalith), ends 2026-07-09, annual cost \$25,629.50 [S:contracts#269]

- Physical Therapy Assistant (Bubblebox), ends 2026-07-11, annual cost \$43,489.19 [S:contracts#828]

- Software Test Engineer III (Tazzy), ends 2026-07-16, annual cost \$794,355.18 [S:contracts#75]

- Community Outreach Specialist (Twitterwire), ends 2026-07-18, annual cost \$429,560.04 [S:contracts#28]

- Senior Financial Analyst (Quimm), ends 2026-07-20, annual cost \$617,931.33 [S:contracts#479]

- Chief Design Engineer (Avaveo), ends 2026-07-21, annual cost \$134,810.51 [S:contracts#951]

- Tax Accountant (Eamia), ends 2026-07-23, annual cost \$239,805.09 [S:contracts#335]

- Budget/Accounting Analyst I (Edgeblab), ends 2026-07-24, annual cost \$438,402.64 [S:contracts#122]

- Database Administrator III (Minyx), ends 2026-07-25, annual cost \$537,232.31 [S:contracts#249]

- Senior Quality Engineer (Skipstorm), ends 2026-07-26, annual cost \$870,326.32 [S:contracts#278]

- Assistant Media Planner (Browsecat), ends 2026-07-29, annual cost \$602,321.73 [S:contracts#477]

- Food Chemist (Yacero), ends 2026-08-03, annual cost \$74,826.93 [S:contracts#369]

- Speech Pathologist (Mynte), ends 2026-08-04, annual cost \$837,154.01 [S:contracts#992]

- Programmer Analyst IV (Fliprune), ends 2026-08-05, annual cost \$313,027.85 [S:contracts#352]

- Quality Control Specialist (Viva), ends 2026-08-05, annual cost \$738,046.00 [S:contracts#993]

- VP Marketing (Miboo), ends 2026-08-08, annual cost \$467,586.25 [S:contracts#814]

- Nurse (Brainsphere), ends 2026-08-09, annual cost \$779,400.18 [S:contracts#94]

- Legal Assistant (Fliprune), ends 2026-08-09, annual cost \$714,735.42 [S:contracts#314]

- Systems Administrator IV (Podcat), ends 2026-08-13, annual cost \$164,494.43 [S:contracts#524]

- Database Administrator III (Blogspan), ends 2026-08-14, annual cost \$909,527.45 [S:contracts#280]

- Chief Design Engineer (Trilith), ends 2026-08-17, annual cost \$254,314.46 [S:contracts#616]

- Software Consultant (Bubblebox), ends 2026-08-19, annual cost \$901,709.42 [S:contracts#86]

- Software Consultant (Realpoint), ends 2026-08-20, annual cost \$418,487.76 [S:contracts#481]

- Human Resources Assistant IV (Podcat), ends 2026-09-02, annual cost \$893,297.00 [S:contracts#488]

- Office Assistant II (Bubbltube), ends 2026-09-03, annual cost \$522,306.96 [S:contracts#672]

Penalties defined in these contracts: not available in the data. The structured evidence contains no penalty terms, and no document evidence was retrieved.

✓ Grounded — every cited source resolves to retrieved evidence (validateAnswer passed).

SOURCES — 38 ROWS • 0 DOCUMENT CHUNKS

[S:contracts#760]

CONTRACTS • ROW 760

contract_id: Dental Hygienist • vendor: Edgepulse • end_date: 6/11/2026 • annual_cost: 779823.65

[S:contracts#737]

CONTRACTS • ROW 737

contract_id: Paralegal • vendor: Voomm • end_date: 6/11/2026 • annual_cost: 95103.45

[S:contracts#523]

CONTRACTS • ROW 523

contract_id: Geological Engineer • vendor: Realbuzz • end_date: 6/12/2026 • annual_cost: 133353.76

[S:contracts#999]

CONTRACTS • ROW 99

contract_id: Quality Engineer • vendor: Brainsphere • end_date: 6/17/2026 • annual_cost: 844932.35

[S:contracts#673]

CONTRACTS • ROW 673

contract_id: Occupational Therapist • vendor: Fanoodle • end_date: 6/17/2026 • annual_cost: 132126.71

Show all 38 rows

Ask another question

Aletheia • grounded knowledge assistant • source • routing • hybrid SQL • RAG • cited & verified

Contract Intelligence — 38 expiring, \$18.9M, cited rows, honest on penalties

Full guide: [contract-intelligence/user-guide.md](https://contract-intelligence.com/user-guide.md).

2. Case File Q&A — page-cited findings, corroboration, and surfaced conflicts

Ask "What was the final child support amount, and who got primary residence?" → **\$1,285/month + primary residence to Joni Carter**, cited to **Page 24 (Final Judgment)**. Ask about the grounds → **corroborated across both documents**. Ask the filing date → it **surfaces the conflict** (the cover sheet says 10 Feb, the narratives say 3 Feb) instead of silently picking one.

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When did Joni Carter file for divorce?

ASK

ROUTING DECISION

question → Documents · RAG

The question is about a specific event in the Carter family-court case, which is in the documents source.

ANSWER

Joni Carter filed for divorce on **February 3, 2026** [P:family-court#6] . This date is corroborated by the background narrative, which also states she filed on February 3, 2026 [P:carter-story#2] .

Note: The cover sheet of the court file lists a "Filed" date of **February 10, 2026** [P:family-court#1] . This conflicts with the February 3, 2026 date stated in the narrative and case summary. The evidence provides two different dates for the filing, and this conflict cannot be resolved from the available sources.

✓ Grounded – every cited source resolves to retrieved evidence (validateAnswer passed).

SOURCES – 8 ROWS · 8 DOCUMENT CHUNKS

[P:carter-story#1]

CARTER-STORY · PAGE 1
CASE BACKGROUND – REASON FOR DIVORCE The Carter Family Story (Fictional) Timeline of Events Leading to Divorce ● 2014 – Marriage and New Beginnings Joni Carter and Michel Carter married on June 14, 2014, in Phoenix, Arizona. They –

[P:family-court#1]

FAMILY-COURT · PAGE 1
PAGE 1 – COVER SHEET In the Matter of: Joni Carter (Petitioner) vs. Michel Carter (Respondent) Type: Dissolution of Marriage Filed: 10 February 2026 Judge: Hon. Rebecca Lawson □

[P:family-court#3]

FAMILY-COURT · PAGE 3
PAGE 3 – CASE SUMMARY This case involves the dissolution of a 12-year marriage between Joni Carter and Michel Carter, including disputes over custody, financial assets, and allegations of addiction-related instability affecting the family e...

[P:family-court#6]

FAMILY-COURT · PAGE 6
ularly. During a financial argument, a physical escalation occurred in which Michel raised his hand toward Joni, causing fear and immediate distress. No severe physical injury occurred, but the event significantly impacted family safety pe...

[P:family-court#24]

FAMILY-COURT · PAGE 24
PAGE 24 – FINAL JUDGMENT ● Marriage dissolved ● Joint custody granted ● Primary residence: Joni Carter ● Asset division: equal split ● Child support: \$1,285/month ● Home sale ordered within 12 months □ EXH...

[P:carter-story#2]

CARTER-STORY · PAGE 2
emained unstable. ● Decision to File for Divorce On February 3, 2026, after months of deliberation and legal consultation, Joni filed for divorce. The decision was motivated by: 1. Chronic gambling and addiction, desp...

[P:family-court#6]

FAMILY-COURT · PAGE 6
PAGE 6 – CHILDREN INFORMATION ● Emma Carter – 11 ● Noah Carter – 8 ● Olivia Carter – 5 All children reside in Phoenix, Arizona and attend local schools. □ PAGES 7-10 – DETAILED MARITAL BACKGROUND The parties married in 2014 in Ph...

[P:carter-story#3]

CARTER-STORY · PAGE 3
ambling websites: various online platforms with international transactions ● Counseling centers: Phoenix Recovery Center, Desert Horizon Therapy ● Schools: Desert Ridge Elementary, Phoenix Pre-K Program Summary of Divorce Rea...

Ask another question

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Case File Q&A — the filing-date conflict surfaced with both citations

Full guide: [case-file-qa/user-guide.md](#).

3. Maintenance Spend Intelligence — cited spend, and honest refusal over fiction

Ask "How much did we spend on maintenance in 2026, and which vendors cost the most?" → **\$13,485.66 across 248 tickets** (2026), **\$40,597.00 all-time**, top vendors cited and drillable. Ask the literal "which customers have overdue payments and what does the agreement say about suspension?" → it **honestly refuses**: this data has no payment-status field and no service agreement, so it explains why and **pivots to the spend analysis it can do** — rather than inventing an overdue list.

Aletheia.

KNOWLEDGE ASSISTANT

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Which customers have overdue payments and what does the agreement say about service suspe

ASK

ROUTING DECISION

question → **Structured** · SQL

The question asks about overdue payments and service suspension terms, which are business operations questions; the structured maintenance_spend intent will honestly report that the data has no such fields, and the documents contain no business agreement.

intents: maintenance_spend

ANSWER

The maintenance data contains no payment-status, paid/unpaid, due-date, or service-suspension field, and there is no service-agreement document. The columns that exist are: Ticket ID, Vendor, Invoice, Labor Cost, Parts Cost, Total Cost, and Completion Date. The vendors are maintenance providers the school pays – they are not customers who owe money. Therefore, I cannot identify which customers have overdue payments or state what the agreement says about service suspension from this data.

However, the total maintenance spend is \$40,597.00 across 750 tickets. I can break this down by vendor or year if needed.

Verified figures (computed over the cited maintenance rows):

- total maintenance spend is \$40,597.00 across 750 tickets **[S:maintenance#5]**.

- top vendor by spend is Oyoba at \$949.94 across 3 tickets **[S:maintenance#5]**.

✓ Grounded – every cited source resolves to retrieved evidence (validateAnswer passed).

SOURCES – 15 ROWS · 8 DOCUMENT CHUNKS

[S:maintenance#5]	MAINTENANCE · ROW 5 ticket_id: Cookies · vendor: Oyoba · total_cost: 549.98 · completion_date: 3/7/2024
[S:maintenance#685]	MAINTENANCE · ROW 685 ticket_id: Juice · vendor: Zoovu · total_cost: 499.98 · completion_date: 10/31/2026
[S:maintenance#257]	MAINTENANCE · ROW 257 ticket_id: Jigsaw Puzzles · vendor: Voolith · total_cost: 412.98 · completion_date: 8/18/2024
[S:maintenance#234]	MAINTENANCE · ROW 234 ticket_id: Specialty Salts · vendor: Kayveo · total_cost: 399.98 · completion_date: 2/7/2026
[S:maintenance#471]	MAINTENANCE · ROW 471 ticket_id: Poultry · vendor: Dabtype · total_cost: 389.98 · completion_date: 2/8/2025

▼ Show all 15 rows

← Ask another question

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Maintenance — honest refusal of the overdue question, pivoting to real spend

The trust property (why this isn't naive RAG)

The single thing that distinguishes a real knowledge assistant from "PDFs in a vector DB" is that it is **honest about what it doesn't know**. This MVP makes that a tested guarantee, not a hope:

- **Every factual claim cites a resolvable source** (a SQLite row id or a PDF page). An answer that makes an uncited claim is **rejected** by `validateAnswer()`.
- **It states absence instead of fabricating** — no penalty data → "not available"; no payment-status field → "I can't determine overdue payments, and here's why"; a source conflict → both values surfaced, not silently resolved.
- **It never invents a join** between unrelated sources (your school operations data and the Carter case file share no key — they're composed and cited separately, never merged).

These behaviors are enforced as **red automated tests** (per-feature content-fidelity gates + journey tests that run against the live deployment), not described in prose.

Data Quality Assessment

This is a selling point, not a footnote. Your stated fear is naive work — "someone who simply uploads PDFs into a vector database and performs semantic search." The antidote is **source-vetting judgment**: knowing what your data can and cannot honestly answer before building. We inspected **all 9 provided sources at the row level**, built features only where a source can ground a real, cited answer, and **deliberately dropped the five that can't** — naming the exact defect for each.

4 of 9 sources support real, grounded features. 5 were vetted and dropped — for specific, named defects, not for time.

#	Source	Verdict	The defect that decides it
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1	school data 1.csv — contracts	✓ Built on	Clean costs/dates → Contract Intelligence. (Contract ID actually holds job titles; some End<Start; no penalty column — handled honestly.)
2	school data 3.csv — maintenance	✓ Built on	Clean spend → Maintenance Spend. No payment-status/due-date field → overdue questions honestly refused, not faked.
3	Family Court Case File (PDF)	✓ Built on	Page-numbered, citable → Case File Q&A (Page-24 Final Judgment).
4	Carter Story (PDF)	✓ Built on	Corroborating narrative for the case file.
5	school data 2.csv — enrollment	✗ Dropped	term_name is 100% a Ruby error string; status holds gender values — columns are the wrong data entirely .

#	Source	Verdict	The defect that decides it
6	school data 4.csv — payroll	⚠ Deferred	Pay-method/notes columns are 100% error strings; the headline fields are corrupt.
7	school data 6.csv — payroll (alt)	⚠ Deferred	pay_month ranges 1–100; payment method is a random integer; can't reconcile with #6.
8	school data 5.csv — invoice totals	✗ Dropped	All 788 rows are identical (180/6/1080) — zero variance to analyze or cite.
9	school data .csv — person list	✗ Out of scope	Generic directory (incl. PII-shaped fields); no spec question touches it.

****That discipline is the product**** — the system says "I can't answer that from this data" (the maintenance-overdue and contract-penalty cases) rather than fabricate, applied not just at answer time but at **data-intake time**.

Full row-level evidence — [docs/product/data-quality-assessment.md](#) (every source, the exact defect, and what clean re-supply would un-block).

Extensible by design

The router + retrieval layer is **source-pluggable**. Your future integrations — CRM, email, cloud storage, case management — each register as a new retrieval source behind the same router contract, **without changing** the grounding/citation/validation layer. The per-fact citation model is what makes per-source access control addable later.

English + Hebrew

The demo answers in **both English and Hebrew today** (the embedding model is multilingual; citations are preserved across languages). Each feature's guide includes a Hebrew golden example. RTL UI polish and Hebrew-tuned prompts are the natural next step — the architecture is already ready for them.

Stack

Next.js · DeepSeek (deepseek-chat) for routing + generation · local Xenova/multilingual-e5-small embeddings (no embeddings key; the Hebrew seam) · bundled read-only SQLite + an in-app vector index · deployed on Vercel. Self-contained and reproducible (npm run build:index rebuilds the index deterministically from the source data).

Quality floor: every push runs CI — doc-lint + doc-structure-lint + typecheck + 40 unit tests; 13 journey tests run against the live deployment. The content-fidelity gates above are part of that floor.